

Path-level Theorem B: existence of a bigrading-preserving injection, and where canonicity breaks

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Abstract

The companion writeup `2026-05-06-theorem-B-multiset.tex` proves Theorem B at the multiset level: $M_{(3,2,1)} = M_{(3,1,1)} * \text{atom}_{\text{RTL}}$, where $\text{atom}_{\text{RTL}} = (1, 7, 1)$ is the hidden non-negative multiset $2M_{(3,2)} - M_{(4,1)}$ on the canonical σ_1 -G1 basis B_4 . This note attempts the path-level lift. We prove the *existence* of a bigrading-preserving injection

$$f : \text{Paths}(V_{(3,1,1)}) \times \text{Paths}(V_{(4,1)}) \hookrightarrow \text{Paths}(V_{(3,1,1)}) \times \text{Paths}(V_{(3,2)}) \sqcup \text{Paths}(V_{(3,1,1)}) \times \text{Paths}(V_{(3,2)})$$

whose set-complement has cardinality $M_{(3,2,1)}$ at every bigrade and hence bijects with $\text{Paths}(V_{(3,2,1)})$. The proof reduces to two ingredients: (i) the componentwise inequality $M_{(4,1)} \leq 2M_{(3,2)}$ on B_4 , which makes a bigrading-preserving injection ι at $n = 5$ possible by pigeonhole; (ii) the convolution lemma, which lifts ι to f and identifies the slack as a multiset of cardinality $M_{(3,2,1)}$. We are honest about the limit: the injection is constructive but *not canonical*. At each $|S|$ -level the choice of injection is determined only up to permutation of paths. Pinning down a W -graph-natural injection — one that respects the KL graph or the $S_5 \downarrow S_4$ branching — remains open, as does the path-level interpretation of atom_{RTL} *itself* (Open Question Q3, end of `2026-05-06-theorem-B-multiset.tex`).

1 Setup: closed W -graph paths and the bigrading

We adopt the framework of `2026-04-24-sigma1-G1.tex` (same directory). Let $H_q = H_q(\mathfrak{S}_n)$ be the Iwahori–Hecke algebra over $\mathbb{Z}[v, v^{-1}]$ with $q = v^2$, and for each partition $\lambda \vdash n$ let V_λ be the cell module on the Kazhdan–Lusztig (KL) basis $\{b_w : w \in \mathcal{C}_\lambda\}$, indexed by standard Young tableaux of shape λ . Fix the staircase reduced word

$$\underline{w}_0 = (i_1, \dots, i_N), \quad N = \binom{n}{2}, \quad \underline{w}_0 = 1; 2, 1; 3, 2, 1; \dots; n-1, \dots, 1,$$

and the Bott–Samelson operator $\Pi_q := \prod_{j=1}^N (T_{i_j} + 1)$. The Apr 24 Lemma decomposes each factor in the KL basis as

$$T_i + 1 = (1 + q) D_i + v M_i,$$

with D_i the diagonal $\{0, 1\}$ -matrix recording left descents ($s_i \cdot w < w$) and M_i a non-negative off-diagonal matrix whose entries are KL μ -coefficients; both have entries in $\mathbb{Z}_{\geq 0}$, and M_i flips the length-parity of vertices.

Closed W -graph paths

Expanding the product:

$$\Pi_q = \sum_{S \subseteq \{1, \dots, N\}} (1+q)^{|S|} v^{|S^c|} \prod_{j=1}^N X_j^S, \quad X_j^S = \begin{cases} D_{i_j} & j \in S, \\ M_{i_j} & j \notin S, \end{cases} \quad (1)$$

where the inner product respects the order $j = 1, 2, \dots, N$. The trace on V_λ becomes

$$\sigma_1(V_\lambda) := \text{tr}(\Pi_q|_{V_\lambda}) = \sum_{S \subseteq \{1, \dots, N\}} (1+q)^{|S|} v^{|S^c|} \cdot \text{tr}\left(\prod_j X_j^S\right).$$

Since each X_j^S has non-negative entries, every $\text{tr}(\prod_j X_j^S) \in \mathbb{Z}_{\geq 0}$. By the parity vanishing of KL coefficients (Apr 24, Lemma 2(ii)), the trace vanishes unless $|S^c|$ is even, so $v^{|S^c|} = q^{|S^c|/2}$ is an honest non-negative power of q .

Definition 1 (Closed W -graph path). A *closed W -graph path* in V_λ is a tuple $\pi = (w_0, w_1, \dots, w_N; S; \mathbf{e})$ with

- $w_0, \dots, w_N \in \mathcal{C}_\lambda$, $w_0 = w_N$ (closed);
- $S \subseteq \{1, \dots, N\}$;
- for each $j \in S$, $w_{j-1} = w_j$, and s_{i_j} is a left descent of w_{j-1} (i.e., $(D_{i_j})_{w_{j-1}, w_{j-1}} = 1$);
- for each $j \notin S$, $\mathbf{e}_j$ is a choice of edge in the W -graph from w_{j-1} to w_j with positive multiplicity $(M_{i_j})_{w_j, w_{j-1}}$, counted with that multiplicity.

The *bigrading* of π is the pair (c, d) with

$$c := |S^c|/2, \quad d := |S|, \quad 2c + d = N.$$

The set of all closed W -graph paths in V_λ , weighted with multiplicity (each $j \notin S$ contributing $(M_{i_j})_{w_j, w_{j-1}}$ many distinct edge-choices), is denoted $\text{Paths}(V_\lambda)$.

The trace formula (1) translates into the path generating identity

$$\sigma_1(V_\lambda) = \sum_{\pi \in \text{Paths}(V_\lambda)} (1+q)^{d(\pi)} q^{c(\pi)}. \quad (2)$$

Equivalently, for $k = 0, 1, \dots, N$ with $k \equiv N \pmod{2}$, write

$$n_k(\lambda) := |\{\pi \in \text{Paths}(V_\lambda) : d(\pi) = k\}|,$$

so $\sigma_1(V_\lambda) = \sum_k n_k(\lambda) (1+q)^k q^{(N-k)/2}$ recovers Theorem 1 of 2026-04-24-sigma1-G1.tex.

Reduction to the post-prefix basis B_{2m}

Each $\sigma_1(V_\lambda)$ has a maximal $q^{a_\lambda} (1+q)^{b_\lambda}$ prefix; dividing through gives the core $A_\lambda(q)$ in the post-prefix basis B_{2m_λ} where $m_\lambda = (\deg A_\lambda)/2$. Concretely, if π has bigrade (c, d) with $d \geq b_\lambda$ and $c \geq a_\lambda$, then π contributes weight $q^{c-a_\lambda} (1+q)^{d-b_\lambda}$ to A_λ . Define the *post-prefix bigrading* on $\text{Paths}(V_\lambda)$ by

$$(\bar{c}, \bar{d}) := (c - a_\lambda, d - b_\lambda).$$

By construction $A_\lambda(q) = \sum_\pi q^{\bar{c}(\pi)}(1+q)^{\bar{d}(\pi)}$, and the B_{2m_λ} -multiplicity vector $M_\lambda = (m_0^{(\lambda)}, \dots, m_{m_\lambda}^{(\lambda)})$ satisfies

$$m_{\bar{c}}^{(\lambda)} = |\{\pi \in \text{Paths}(V_\lambda) : \bar{c}(\pi) = \bar{c}\}|.$$

For the rest of this paper, “bigrading” refers to the post-prefix $(\bar{c}, \bar{d})$, and we drop the bars. The five multiplicity vectors at hand (recomputed in `2026-05-06-theorem-B-multiset.tex`, equation (5)) are

$$\begin{aligned} M_{(3,2,1)} &= (1, 9, 15, 2), & M_{(5,1)} &= (1, 3, 9, 14), \\ M_{(3,1,1)} &= (1, 2), & & \\ M_{(3,2)} &= (1, 5, 3), & M_{(4,1)} &= (1, 3, 5). \end{aligned} \tag{3}$$

The convolution lemma (Apr 24, Lemma 3.1 of `2026-05-06-twin-multiset.tex`) gives that on Cartesian products of paths, the bigrading $\bar{c}$ adds, and multiplication of the polynomial atoms is convolution of multiplicity vectors.

2 The $n = 5$ injection ι

Theorem 2 (ι at $n = 5$). *There exists a bigrading-preserving injection of multisets*

$$\iota : \text{Paths}(V_{(4,1)}) \hookrightarrow \text{Paths}(V_{(3,2)}) \sqcup \text{Paths}(V_{(3,2)}),$$

i.e., for each $c \in \{0, 1, 2\}$, ι injects the $\bar{c} = c$ subset of $\text{Paths}(V_{(4,1)})$ into the $\bar{c} = c$ subset of two disjoint copies of $\text{Paths}(V_{(3,2)})$. The set-complement of ι has multiplicity vector $\text{atom}_{\text{RTL}} = (1, 7, 1)$ on B_4 .

Proof. Existence reduces to the bigrading-by-bigrading inequality

$$|\text{Paths}(V_{(4,1)})_c| \leq 2 \cdot |\text{Paths}(V_{(3,2)})_c| \quad \text{for each } c \in \{0, 1, 2\},$$

which by Definition 1 and (3) is precisely the componentwise inequality

$$M_{(4,1)} \leq 2 M_{(3,2)} \iff (1, 3, 5) \leq (2, 10, 6).$$

This is checked entry by entry: $1 \leq 2$, $3 \leq 10$, $5 \leq 6$. All three inequalities hold (the last by a hair — 1 slot of slack).

By pigeonhole on each c -fibre, an injection ι from $\text{Paths}(V_{(4,1)})_c$ into $\text{Paths}(V_{(3,2)})_c \sqcup \text{Paths}(V_{(3,2)})_c$ exists. To make a specific choice, fix any total order on $\text{Paths}(V_{(3,2)})_c$ (for example, lex order on the data $(w_0, S, \mathbf{e})$ of Definition 1); inject the $\bar{c} = c$ paths of $V_{(4,1)}$ in any order into the first $|M_{(4,1)}|_c$ slots of (copy 1) $\sqcup$ (copy 2), ordered first-copy-first.

The complement at each c has cardinality $2|M_{(3,2)}|_c - |M_{(4,1)}|_c$, which by Lemma 2.1 of `2026-05-06-theorem-B-` equals

$$2M_{(3,2)} - M_{(4,1)} = (1, 7, 1) = \text{atom}_{\text{RTL}}. \quad \square$$

Remark 3 (The slack at $c = 2$ is one slot). The injection is barely possible at $c = 2$: $|M_{(4,1)}|_2 = 5$ and $2|M_{(3,2)}|_2 = 6$, so the slack is just one. Once that single $c = 2$ slack-element is chosen, the rest of the $c = 2$ injection is forced (every other path in the doubled $V_{(3,2)}$ at $c = 2$ must receive a path from $V_{(4,1)}$). Conversely, at $c = 1$ there are 7 slack slots out of 10, so the choice of injection is far from determined. This asymmetry — single slack at the boundary, abundant slack in the middle — is exactly the palindromic shape $(1, 7, 1)$ of atom_{RTL} .

3 Lifting ι to f at $n = 6$

Theorem 4 (Path-level Theorem B, B1 form). *There exists a bigrading-preserving injection of multisets*

$$f : \text{Paths}(V_{(3,1,1)}) \times \text{Paths}(V_{(4,1)}) \hookrightarrow \text{Paths}(V_{(3,1,1)}) \times \text{Paths}(V_{(3,2)}) \sqcup \text{Paths}(V_{(3,1,1)}) \times \text{Paths}(V_{(3,2)})$$

whose set-complement has multiplicity vector $M_{(3,2,1)} = (1, 9, 15, 2)$ on B_6 and is therefore in bijection with $\text{Paths}(V_{(3,2,1)})$.

Proof. Define $f := \text{id}_{\text{Paths}(V_{(3,1,1)})} \times \iota$, where ι is the $n = 5$ injection of Theorem 2. Concretely, f sends a pair (α, β) with $\alpha \in \text{Paths}(V_{(3,1,1)})$ and $\beta \in \text{Paths}(V_{(4,1)})$ to the pair $(\alpha, \iota(\beta))$, where $\iota(\beta)$ lies in one of the two copies of $\text{Paths}(V_{(3,2)})$.

Bigrading preservation. The bigrading on a Cartesian product of paths adds: $\bar{c}(\alpha, \beta) = \bar{c}(\alpha) + \bar{c}(\beta)$ (and similarly for $\bar{d}$, by (2) and the convolution lemma). Since ι preserves $\bar{c}$ on $V_{(4,1)} \rightarrow V_{(3,2)} \sqcup V_{(3,2)}$, $\bar{c}(\alpha, \beta) = \bar{c}(\alpha) + \bar{c}(\beta) = \bar{c}(\alpha) + \bar{c}(\iota(\beta)) = \bar{c}(\alpha, \iota(\beta))$.

Injectivity. If $(\alpha_1, \iota(\beta_1)) = (\alpha_2, \iota(\beta_2))$, then $\alpha_1 = \alpha_2$ and $\iota(\beta_1) = \iota(\beta_2)$. By injectivity of ι , $\beta_1 = \beta_2$. So f is injective.

Cardinality of the complement. At each $\bar{c} = c$ on B_6 :

$$\begin{aligned} |\text{codomain}|_c &= 2|\text{Paths}(V_{(3,1,1)}) \times \text{Paths}(V_{(3,2)})|_c \\ &= 2(M_{(3,1,1)} * M_{(3,2)})_c, \\ |\text{image of } f|_c &= |\text{Paths}(V_{(3,1,1)}) \times \text{Paths}(V_{(4,1)})|_c \\ &= (M_{(3,1,1)} * M_{(4,1)})_c, \\ |\text{complement}|_c &= 2(M_{(3,1,1)} * M_{(3,2)})_c - (M_{(3,1,1)} * M_{(4,1)})_c \\ &= (M_{(3,1,1)} * (2M_{(3,2)} - M_{(4,1)}))_c \\ &= (M_{(3,1,1)} * \text{atom}_{\text{RTL}})_c. \end{aligned}$$

By the multiset Theorem B1 (Theorem 3.1 of 2026-05-06-theorem-B-multiset.tex), $M_{(3,1,1)} * \text{atom}_{\text{RTL}} = M_{(3,2,1)}$. Hence the complement at each c has cardinality $|M_{(3,2,1)}|_c = |\text{Paths}(V_{(3,2,1)})|_c$.

Bijection with $\text{Paths}(V_{(3,2,1)})$. Pick, at each $c \in \{0, 1, 2, 3\}$, any bijection between the complement at level c and $\text{Paths}(V_{(3,2,1)})_c$ (e.g., lex order both sides). The resulting bijection is by construction bigrading-preserving. $\square$

Corollary 5 (Path-level Theorem B, B2 form). *There exists a bigrading-preserving injection*

$$f' : \text{Paths}(V_{(3,1,1)}) \times \text{Paths}(V_{(3,2)}) \hookrightarrow \text{Paths}(V_{(3,1,1)}) \times \text{Paths}(V_{(4,1)}) \sqcup \text{Paths}(V_{(3,1,1)}) \times \text{Paths}(V_{(4,1)})$$

whose set-complement has multiplicity vector $M_{(5,1)} = (1, 3, 9, 14)$ and bijects with $\text{Paths}(V_{(5,1)})$.

Proof. Repeat Section 2 with $(3, 2)$ and $(4, 1)$ swapped: by (3),

$$M_{(3,2)} \leq 2M_{(4,1)} \iff (1, 5, 3) \leq (2, 6, 10),$$

which holds entry by entry (each inequality strict, in fact). So an injection ι' at $n = 5$ exists with slack $2M_{(4,1)} - M_{(3,2)} = (1, 1, 7) = \text{atom}_{\text{LTR}}$. Repeating the argument of Theorem 4 with ι' in place of ι and using the multiset Theorem B2 ($M_{(3,1,1)} * \text{atom}_{\text{LTR}} = M_{(5,1)}$), the complement of f' has cardinality $M_{(5,1)}$ and bijects with $\text{Paths}(V_{(5,1)})$. $\square$

4 What is and is not canonical

What the proof gives

Theorem 4 is a genuine path-level statement: an explicit (constructive) injection of finite multisets, whose complement is in bijection with $\text{Paths}(V_{(3,2,1)})$. The proof is concrete — given a fixed enumeration of paths, the injection can be tabulated. It is correct, it is checkable, and it lifts the multiset Theorem B from polynomial arithmetic to a finite-set statement.

What the proof does not give

The injection f is canonical only up to two layers of arbitrary choice:

1. **The $n = 5$ injection ι .** At $c = 1$, only 3 of 10 slots in the doubled $V_{(3,2)}$ receive paths from $V_{(4,1)}$. Which 3 is unspecified. The proof works for any choice. (At $c = 0$ and $c = 2$ the choice is much tighter — 1 of 2, and 5 of 6, respectively — but still not unique.)
2. **The complement-to- $P_{(3,2,1)}$ bijection.** At each c on B_6 , we pick any bijection between two finite sets of equal cardinality. The choice is unspecified.

The natural question is: *is there a canonical, W -graph-theoretic, choice that picks out f uniquely?* A canonical ι at $n = 5$ would presumably exploit one of:

- the $S_5 \downarrow S_4$ branching, since both $V_{(3,2)} \downarrow$ and $V_{(4,1)} \downarrow$ contain $V_{(3,1)}$ as a common summand,
- the S_5 -Mullineux involution (or its σ -twist), since the pair $(M_{(3,2)}, M_{(4,1)})$ has the suggestive form $(1, 5, 3)$ and $(1, 3, 5)$ — a palindromic “shoulder swap” between 5 and 3 in positions 1 and 2,
- or a path-level operator on the W -graph that “absorbs” the second copy of $V_{(3,2)}$ into $V_{(4,1)}$.

None of these has been pinned down. They are open.

The structural meaning of atom_{RTL} remains open

Theorem 2 produces the slack as a multiset of cardinality $(1, 7, 1)$, but does *not* identify it. Open Question Q3 (preserved from 2026-05-06-theorem-B-multiset.tex) asks: is the slack of the canonical ι in bijection with the closed W -graph paths in $V_{(3,2,1)}$ that traverse the s_5 -coloured edge? An affirmative answer would simultaneously

- give atom_{RTL} a structural identity,
- and pin down the canonical injection ι .

Verifying this would require enumerating closed W -graph paths in $V_{(3,2,1)}$ and counting those that touch s_5 at each c , then comparing with $(1, 7, 1)$. The expected counts would be (working backwards through the convolution): at $c = 0$, one path; at $c = 1$, seven paths; at $c = 2$, one path. Both the counts and the identification of which paths touch s_5 are computable.

Quantitative “why” for $2M_{(3,2)} \geq M_{(4,1)}$

The componentwise inequality

$$2m_c^{(3,2)} \geq m_c^{(4,1)} \quad (c \in \{0, 1, 2\})$$

is checked numerically in this paper but is not derived structurally. A path-level proof would presumably exhibit, for each closed path $\beta \in V_{(4,1)}$ at bigrade c , a pair (or perhaps, with finer

accounting, two paths) of closed paths in $V_{(3,2)}$ at the same c . Such a structural “fattening” map $\beta \mapsto (\alpha_1, \alpha_2)$ would simultaneously prove the inequality and pin down ι . We have not constructed it.

Computational check

All cardinalities used in this paper come from (3), themselves computed in `~/projects/scratch/2026-05-06-theo` (May 6, 2026) and re-verified against `~/projects/scratch/2026-05-06-twin-wgraph/decompose_output.txt`. The convolution $M_{(3,1,1)} * \text{atom}_{\text{RTL}} = (1, 2) * (1, 7, 1) = (1, 9, 15, 2) = M_{(3,2,1)}$ is verified directly:

$$(1, 2) * (1, 7, 1) = (1 \cdot 1, 1 \cdot 7 + 2 \cdot 1, 1 \cdot 1 + 2 \cdot 7, 2 \cdot 1) = (1, 9, 15, 2). \quad \checkmark$$

The symmetric convolution $(1, 2) * (1, 1, 7) = (1, 3, 9, 14) = M_{(5,1)}$ is verified analogously.

5 Discussion: this is a real result, but a small one

What this proof contributes beyond `2026-05-06-theorem-B-multiset.tex`: a clean reformulation of Theorem B as a containment-and-complement of finite multisets of W -graph paths, with the slack at $n = 5$ identified as the cardinality vector atom_{RTL} . The convolution lemma then mechanically lifts this to $n = 6$.

What this proof does *not* contribute: a canonical or natural choice of injection. The injection is determined only up to choices at two stages, and at the second stage (the complement-to- $P_{(3,2,1)}$ bijection) we make no attempt at canonicity at all. A path-level proof in the strong sense — one that reads “here is the natural map, and the slack *is* atom_{RTL} ” — requires identifying a W -graph-theoretic feature that distinguishes $(3, 2)$ from $(4, 1)$ at S_5 . The branching $V_{(3,2)} \downarrow = V_{(2,2)} \oplus V_{(3,1)}$ vs $V_{(4,1)} \downarrow = V_{(4)} \oplus V_{(3,1)}$ is a natural candidate: the difference is $V_{(2,2)}$ vs $V_{(4)}$, both small modules at S_4 . Whether the slack $(1, 7, 1)$ admits an interpretation in terms of these is a question for the next session.

The author flags the present result as a partial success: the multiset Theorem B is now lifted to the finite-set level (existence of f), but the structural payoff — a canonical f — is open, as is Open Question Q3 on the path-level identity of atom_{RTL} .